PARTNERSHIP MEMO

United States–Japan Strategic Energy & Industrial Partnership

Where deeper industrial integration creates strategic value, through which instruments, and where to stay out

Generated from Allied Industrial Base Explorer data and a curated external evidence layer (accessed 2026-11-06); evidence, sources and caveats in the appendix. Data as of 30 October 2026.

Executive Decision Summary

BUILD TOGETHER

Permanent Magnets & Drive Systems: Rare-earth oxides & compounds (joint); Rare-earth permanent magnets (Japan->US). Battery Materials & Cells: Natural graphite (joint); Accumulator parts & separators (Japan->US). Solar Manufacturing: Polysilicon (US->Japan). Wind & Hydropower Equipment: Wind turbine generating sets (US->Japan).

TRADE AND STANDARDS

Recognise trusted partner supply of electric vehicles extraction (cobalt ores, lithium minerals, natural graphite); wind turbine generating sets; accumulator parts & separators in procurement and content rules. Common anti-circumvention rules for electric vehicles refining (rare-earth metals, rare-earth oxides & compounds) and rare-earth oxides & compounds. Keep uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores); pgms (iridium, ruthenium); nuclear special materials open to global sourcing.

INSTITUTIONS

Upgrade Japan-US Strategic Investment Initiative (MOU of 4 Sep 2025): sector strategy, reciprocal commitments, a KPI dashboard and an annual review. No new MOU.

STAY OUT

Do not co-subsidise new Automatic controllers capacity. Do not co-subsidise new Gas components capacity. Do not co-subsidise new Semiconductor manufacturing equipment capacity.

Why this partnership matters.

- Of 142 technology x stage cells and electro-industrial nodes assessed for the pair, 35 show an active complementarity (a supplier meeting the other's need, a shared chokepoint, a capacity-constrained trusted supplier or a chain specialisation), 4 direct competition and 79 no strategic case. The relationship is not one of general similarity: most of the electro-industrial base needs no bilateral intervention.[1]
- AIB strategic roles: US Demand & Market Anchor (primary), Coordination / Bridge Hub (primary), Capital & Investment Anchor (primary), Scale Manufacturing & Integration Platform (secondary), Specialized Technology & Manufacturing Anchor (secondary); Japan Capital & Investment Anchor (primary), Specialized Technology & Manufacturing Anchor (primary), Scale Manufacturing & Integration Platform (primary), Coordination / Bridge Hub (secondary).[1]

The central bilateral bargain.

- *Inference.* Japan supplies specialised processed materials and components where the US is short (accumulator parts & separators, rare-earth permanent magnets, wafers, battery materials (foil, electrolyte)); the US supplies polysilicon, wind turbine generating sets, battery recycling feed where Japan is short; and the

two share exposure to the same coercive supplier in rare-earth oxides & compounds and natural graphite, where neither can supply the other and the answer is joint demand, finance and third-country capacity.

- *Inference.* The US brings demand, offtake and capital to Japan's capability; Japan brings specialised capacity and patient finance (JBIC, NEXI, JOGMEC) to the US's market. That is the trade: market access and demand certainty for trusted capacity.

The 4 highest-priority joint bets (tier 1 by the rules in Section 3, one per compact; ranked by urgency then importance):

PRIORITY	SECTOR	NODES IN THE BET (DIRECTION)	STRATEGIC PROBLEM	US CONTRIBUTION	JAPAN CONTRIBUTION	MAIN INSTRUMENT (LEAD NODE)	TIMING
1	Permanent Magnets & Drive Systems	Rare-earth oxides & compounds (joint); Rare-earth permanent magnets (Japan->US)	geopolitical chokepoint	Rare-earth oxides & compounds: its demand for pooled offtake, and finance for non-Chinese supply; Rare-earth permanent magnets: demand and long-term offtake	Rare-earth oxides & compounds: its demand for pooled offtake, and finance for non-Chinese supply; Rare-earth permanent magnets: trusted capacity (9% of world supply)	anchor offtake	12-24 months
2	Battery Materials & Cells	Natural graphite (joint); Accumulator parts & separators (Japan->US)	geopolitical chokepoint	Natural graphite: its demand for pooled offtake, and finance for non-Chinese supply; Accumulator parts & separators: demand and long-term offtake	Natural graphite: its demand for pooled offtake, and finance for non-Chinese supply; Accumulator parts & separators: trusted capacity (11% of world supply)	anchor offtake	12-24 months
3	Solar Manufacturing	Polysilicon (US->Japan)	geopolitical chokepoint	Polysilicon: trusted capacity (31% of world supply)	Polysilicon: demand and long-term offtake	strategic imports	12-24 months
4	Wind & Hydropower Equipment	Wind turbine generating sets (US->Japan)	geopolitical chokepoint	Wind turbine generating sets: trusted capacity (7% of world supply)	Wind turbine generating sets: demand and long-term offtake	anchor offtake	12-24 months

Sources: 1.

- *Inference.* Outside the electro-industrial compacts, the rules also find fuel-trade lanes (oil extraction); they are handled in the trade agenda, not as compacts.

Do not prioritise (Section 10):

- *Recommendation.* Do not co-subsidise new Automatic controllers capacity: both are strong suppliers (US 16%, Japan 11%). Trusted supply is adequate: this is commercial competition.
- *Recommendation.* Do not co-subsidise new Gas components capacity: both are strong suppliers (US 21%, Japan 4%). Trusted supply is adequate: this is commercial competition.
- *Recommendation.* Do not co-subsidise new Semiconductor manufacturing equipment capacity: both are strong suppliers (US 16%, Japan 21%). Trusted supply is adequate: this is commercial competition.

Existing mechanisms and the MEA question (Section 8):

- *Inference.* MEA assessment: **upgrade an existing framework (USJ-SII)**. Gaps: no cross-sector KPI dashboard or review cycle; no bilateral mechanism covers SC-SOLAR, SC-GRID; USJ-SII has a pipeline and finance but no reciprocal commitments or sector strategy.

- *Recommendation.* Do not sign a new energy MOU. Give USJ-SII a sector strategy, reciprocal commitments and a KPI dashboard covering the recommended compacts, reviewed every 12 months.

1. Current State of the Relationship

This section sets out what already exists before anything new is proposed. Facts are from the curated evidence layer (accessed 2026-11-06); items marked partial or unverified rest on secondary sources and should be confirmed before citation.

Alliance and plurilateral architecture

- Japan's memberships in OSI's economic-security agreements table: G7, IPEF, MSP.[1]
- Both sides in 2024 said the 2022-launched Economic 2+2 works as an effective economic-security framework and agreed to hold it periodically; no 2026 meeting found (Feb 2026 Rubio-Motegi and Ichikawa-Rubio contacts only).[2]
- Quad foreign ministers adopted a framework to mobilise up to $20bn for secure critical-mineral supply chains; a separate July 2025 Quad initiative is mentioned only in passing and not sourced.[3]

Trade architecture and frictions

- In the OSI energy basket, US imported US$41.9bn from Japan and Japan imported US$20.1bn from US (bilateral flows 2024, summed over the technology x stage cells).[1]
- Lighthizer and Ambassador Sugiyama signed the Trade Agreement and the Digital Trade Agreement on 7 Oct 2019.[4]
- Signed 28 Mar 2023 and in force on signature without congressional approval; covers cobalt/graphite/lithium/manganese/nickel; no export duties; consult on non-market measures and forced-labour imports; Treasury/USTR treated it as an FTA for the IRA EV credit critical-minerals test.[5]
- Japan government text states that for any Section 232 tariff on pharmaceuticals and semiconductors (incl. SME) the US intends to apply to Japan a rate no greater than that applied to any other country; search snippet only, full document not read.[6]
- EO signed 4 Sep 2025 sets 15% all-in tariff on Japanese goods (reciprocal) and a Section 232 auto/auto-parts rate that tops MFN up to 15% (added rate zero at or above 15%).[7]
- CRS (Apr 2026) describes the DTA as an executive agreement enacted without changes to US law.[8]
- IRS says the new clean vehicle credit (30D) is not allowed for vehicles acquired after 30 Sep 2025, so the CMA's IRA-credit eligibility function is largely moot; no source found on any formal CMA amendment or termination (status in force but unreviewed).[9]
- In Feb 2026 Akazawa said the sides were far apart on first projects; Lutnick claimed Trump has complete discretion while Akazawa said selection follows the investment and consultation committees; resolved only by the 17 Feb announcement; later disputes not found.[10]
- On 20 Feb 2026 the Court ruled 6-3 that IEEPA does not authorise tariffs, removing the legal basis of the 15% Japan reciprocal rate; a temporary Section 122 global surcharge replaced it; Section 232 duties (autos etc.) unaffected.[11]
- Axios (30 Jun 2026) says Nippon reaffirmed an $11bn investment pledge with allocation of $7.8bn unspecified and mostly still promised, the golden share has not been used on management decisions, and leverage rose with an S&P downgrade; the deal closed 18 Jun 2025.[12]
- Jiji reports the US imposed new 12.5% Section 301 tariffs on Japan and dozens of partners after the Section 122 surcharge expired 24 Jul 2026; Japan called it regrettable and the 15% auto rate is said to be maintained; a separate excess-capacity 301 probe (launched Mar 2026) includes Japan. Check USTR Federal Register notice for final rate.[13]

- As of Sep 2026 Akazawa declined detail on a third tranche, with AI and semiconductor projects reported as leading candidates and no date; no third announcement found by 6 Nov 2026.[14]

Industrial and energy agreements

- MOU signed 4 Sep 2025 (Lutnick/Akazawa): presidential investment committee chaired by Commerce Secretary consults a joint consultation committee; US proposes projects; Japan has 45 days to fund each approved project or may decline and forfeit distributions (President may raise tariffs); per-project SPVs split cash 50/50 then 90/10 to the US after a deemed allocation amount; investments before 19 Jan 2029; US Investment Accelerator executes.[15]
- Framework text dated 27 Oct 2025 (signed Tokyo 28 Oct) commits to financing within six months for selected projects (grants/guarantees/loans/equity/offtake/insurance), a ministerial within 180 days, faster permitting, pricing measures to support non-China supply, scrap recycling, a complementary stockpile arrangement, and a Rapid Response Group led by the US Energy Secretary and METI Minister; names no dollar amounts and is non-binding.[16]
- Framework establishes a group led by the US Energy Secretary and Japan's METI Minister to identify priority minerals and accelerate processed-mineral deliveries.[17]
- Greer announced the Action Plan on 19 Mar 2026: the two countries will develop strategic trade policies and border mechanisms to protect downstream industries and lay the foundation for a binding plurilateral agreement supported by price floors; trade press adds deep-sea R&D (not read from primary).[18]
- Signed 28 Mar 2023 and in force on signature without congressional approval; covers cobalt/graphite/lithium/manganese/nickel; no export duties; consult on non-market measures and forced-labour imports; Treasury/USTR treated it as an FTA for the IRA EV credit critical-minerals test.[5]

Investment vehicle, project pipeline and financing

- US government will arrange financing and permitting for at least $80bn of new Westinghouse reactors (mainly AP1000) and receives a participation interest of 20% of distributions above $17.5bn convertible to up to 20% equity; date 28 Oct 2025 per most sources (one says 27 Oct).[19]
- Announced 28 Oct 2025 as preliminary pledges in a launch phase needing review; AA lists nuclear up to $332bn, Mitsubishi Electric AI infrastructure $30bn and Panasonic energy storage $15bn.[20]
- The fact sheet lists up to $100B for Westinghouse AP1000/SMR with possible Japanese supplier involvement (MHI Toshiba IHI); CSIS cites a separate figure of $80B in a US-Westinghouse partnership.[21]
- The fact sheet lists up to $100B for GE Vernova Hitachi BWRX-300 SMR construction with Hitachi GE Vernova as a possible Japanese participant.[22]
- JERA (Sep 2025) and Tokyo Gas (6 Nov 2025) each signed non-binding LOIs for 1 MTPA from Alaska LNG; no binding SPA, JBIC financing or FID found, and it is not among announced SII projects.[23]
- One of the first three funded projects under the $550B vehicle is a $33B 9.2GW SB Energy gas plant in Portsmouth Ohio.[24]
- Commerce lists industrial synthetic diamond manufacturing at about $0.6bn in the first batch announced 17 Feb 2026; the Georgia site and Element Six (De Beers) operator come from secondary reporting only.[25]
- Commerce lists US crude oil export infrastructure at about $2.1bn in the first batch (17 Feb 2026); Sentinel Midstream's Texas GulfLink identification comes from secondary reporting only.[26]
- Commerce lists natural gas generation at about $33.3bn (first batch total about $36bn); the Ohio location and 9.2 GW size come from the Lutnick statement as quoted in secondary coverage.[27]
- ANS (17 Mar 2026) says the 2025 US-Japan agreement projected up to $100bn for Westinghouse reactors but specific projects were not yet identified and Westinghouse is still working transactions.[28]
- Second batch announced 19 Mar 2026 at the Takaichi-Trump summit lists SMRs by GE Vernova Hitachi in Tennessee and Alabama at up to $40bn; TVA Clinch River and Bellefonte/Widows Creek siting is from

secondary reports.[29]

- Commerce lists gas generation in Pennsylvania at up to $17bn with offtake including co-located data centres; NextEra/South Mon 4.3 GW identification is secondary only.[30]
- Commerce lists gas generation in Texas at up to $16bn with offtake including data centres (second batch total about $73bn).[31]
- JBIC signed a loan of about $7m to Japan Invest 1 LLC on 17 Apr 2026 (total co-financing about $23m) with NEXI insuring the private lenders' portion.[32]
- JBIC signed a loan of about $630m to Japan Invest 3 LLC on 17 Apr 2026 (total co-financing about $1.885bn) for a gas generation project integrated with data centre and transmission; NEXI insures private lenders; lenders unnamed.[33]
- JBIC signed about $739m (loan agreement 7 Aug 2026; total co-financing about $2.213bn) to Japan Invest 5 LLC for Texas gas generation under the Initiative; NEXI insures private loans; Citibank Tokyo agent with JPMorgan Tokyo.[34]
- Reported: JBIC US$801m loan to JI4 signed 7 Aug 2026 with NEXI insuring a US$1.6bn private loan, and a Citi-led facility of about US$4.6bn covering JI4 and JI5.[35]

Corporate relationships in the recommended compacts

- Hitachi Energy pledged more than $1B for US grid equipment manufacturing including the $457M South Boston plant.[36]
- Hitachi Energy is reported to commit $528M to a Gallman Mississippi power transformer plant supporting 650+ jobs, with no capacity or timeline released.[37]
- Mitsubishi Electric opened a roughly $200M Memphis Tennessee transformer factory in 2013.[38]
- Mitsubishi Electric announced an $86M advanced switchgear factory in Western Pennsylvania in October 2024.[39]
- AESC paused its $1.6B Florence County plant in June 2025 citing policy and market uncertainty.[40]
- Nippon Steel committed about $11B of new US Steel investment by 2028 (some reports cite $14B) with electrical steel mentioned mainly for automotive.[41]
- Panasonic opened the De Soto plant in July 2025 with about 1100 employees and one of eight lines running versus a 4000-job target.[42]
- Nikkei reported in November 2025 that US Steel would build GOES capability at Big River Osceola Arkansas with mass production in 2028; US Steel confirmed the project to Argus without investment or capacity details (date approximate).[43]
- Toyota began production at its Liberty NC battery plant on 2025-11-12 (reported investment $13.9B with a further $10B pledged); output is mainly hybrid batteries and sources differ on line counts.[44]
- LGES agreed to sell the Ohio JV factory building and related assets to Honda for about $2.85B with the JV continuing via lease and closing targeted for 2026-02-28.[45]
- Panasonic plans to repurpose some US lines including De Soto to data-center energy storage with first cells expected mid-to-late 2028 or early 2029 while EV cell output continues; ramp is slower than planned.[46]
- Hitachi Energy broke ground on 2026-06-29 on a $457M South Boston expansion creating about 825 jobs; the site already employs about 850 and completion date was not stated on the page (other reports say 2028).[47]
- Unverified items are left out of this section; they stay in the evidence register (evidence_register.csv) and the evidence layer.[1]
- *Inference.* The relationship already has two working channels: an investment vehicle with a project pipeline and JBIC/NEXI finance (so far gas power, crude export, synthetic diamonds and SMRs), and a critical-minerals framework with financing, pricing and stockpile commitments but no published project list. What it lacks is a

link between the two and any sector strategy for the electro-industrial chains where the data show complementarity.

2. What Each Country Is Good At — and What It Needs

Cell codes: position / need for **US · Japan**. Position S strong (top quarter of contenders and above the capability floor), M contender, W not a contender; need H high (supply-chain need >= 0.67, or deployment need >= 0.60 at Deployment), L low; - not measured. Source: strategic_position_tbl (engine choices are in country_profiles.csv).

TECHNOLOGY	EXTRACTION	REFINING	COMPONENTS	ASSEMBLY	DEPLOYMENT	RECOVERY
Batteries	W/H · W/H	M/L · M/H	S/L · M/H	S/H · M/H	S/H · S/L	-/H · -/H
Coal	S/L · W/H	M/H · M/H	M/L · S/H	M/L · S/L	S/L · S/L	-/- · -/-
Electric Grid	M/H · W/H	W/H · M/H	M/L · S/L	M/L · M/H	-/L · -/L	-/- · -/-
Electric Vehicles	M/H · W/H	M/H · M/H	S/H · M/L	S/L · M/L	S/L · S/L	-/- · -/-
Gas	S/L · W/H	S/L · W/H	S/L · S/L	M/L · S/L	S/H · S/H	-/- · -/-
Geothermal	-/- · -/-	-/- · -/-	-/- · -/-	-/- · -/-	S/L · S/L	-/- · -/-
Green Hydrogen	M/- · W/-	W/- · W/-	-/- · -/-	-/- · -/-	-/L · -/H	-/- · -/-
Nuclear	W/H · W/L	W/H · W/H	M/L · M/L	M/L · S/L	S/L · S/H	-/- · -/-
Oil	S/L · W/H	S/L · M/L	M/L · M/L	M/L · S/L	S/L · S/L	-/H · -/L
Solar	M/H · W/H	W/H · M/H	S/L · M/L	S/L · M/L	S/L · S/H	-/- · -/-
Wind	S/- · W/-	M/H · W/H	S/L · M/L	S/L · M/H	S/L · S/H	-/- · -/-

Sources: 1.

Electro-industrial nodes where either country has a position or a need (AIB division of labour, 2024 trade; share = world export or production share; need is the node's technology x stage need, inherited, so a proxy; demand = share of allied demand). 75 of 76 nodes:

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

Table A1 in the appendix (75 rows) carries this table in full; the CSV outputs carry every column.

- Revealed policy priority (NIPO / IEA PAMS within-country percentile; policy never enters the classification) is highest for US in: Green Hydrogen assembly (0.95); Green Hydrogen components (0.93); Nuclear components (0.92); Nuclear assembly (0.90); Batteries assembly (0.87); for Japan in: Green Hydrogen assembly (0.99); Green Hydrogen components (0.98); Batteries assembly (0.92); Gas recovery (0.86); Coal assembly (0.84).[1]
- Evidence quality: 42 of 66 cells have high stage-specific evidence for both countries; 19 units are flagged low-evidence.[1]

3. Bilateral Complementarity Map

State counts over all units: a_supplies_b_needs 7; complementary_specialization 8; shared_vulnerability 7; capacity_constrained_trusted_supply 13; direct_competition 4; low_strategic_case 79; deployment_demand 6; not_assessed 18.[1]

Inference. Rules (Section 3 of docs/partnership_memos.md): two strong positions are **direct competition**, never complementarity; a supplier must be strong (or a contender the engine names as the other's trusted partner) and the buyer must have high need, or material demand where trusted capacity is short or a chokepoint exists; **capacity-constrained** where the AIB finds allied capacity short of demand; **shared vulnerability** where both need the node, neither supplies it and a coercive supplier holds at least half of world supply.

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

Table A2 in the appendix (39 rows) carries this table in full; the CSV outputs carry every column.

- Low-priority / open-market nodes (56): synthetic graphite (anode-grade), solar cells, cathode materials & precursors, pv modules & generators, li-ion cells & packs, rare-earth metals, uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores), uranium conversion & enrichment, nuclear reactors & enrichment machinery, fuel assemblies & zirconium, reactor parts, cobalt intermediates & chemicals, lithium chemicals, wind blades & hubs, electric & plug-in cars, pv cover glass, wind towers, lithium minerals, nickel intermediates & chemicals, other accumulators, cobalt ores, rare-earth ores (monazite), ferrite & other magnets, copper smelting & refining, heat pumps & ac units, static converters & inverters, power electronics components, solar thermal water heaters, gallium, germanium & indium, grain-oriented electrical steel (goes), power semiconductors, electric two-wheelers & works trucks, electric motors, insulators & bushings, motor & generator parts, non-oriented electrical steel (ngo), electric buses, trucks & tractors, electricity meters, electric furnaces & heating, integrated circuits, transformers, switchgear & protection, hydro turbine parts, graphite electrodes, blade core wood (balsa), copper ores & concentrates, copper wire & rod, nuclear special materials, overhead conductors, power cables, aluminium wire rod, direct-reduced iron (dri/hbi), pgms (iridium, ruthenium), hydrogen, ammonia & methanol, electric rail vehicles, hydro turbines.[1]
- Deployment cells are demand context, not supply relations: high deployment need in Nuclear (JPN); Green Hydrogen (JPN); Solar (JPN); Wind (JPN); Batteries (USA); Gas (USA and JPN).[1]

4. Trade Agenda

- *Inference.* **What the US should export to Japan:** Polysilicon; Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite); Wind turbine generating sets; Batteries components (accumulator parts & separators); Oil extraction; Wind assembly (electricity meters, overhead conductors, power cables); Battery recycling feed; Batteries assembly (li-ion cells & packs, other accumulators); Gas extraction; Gas refining; Refrigerants; Sensors & controller parts; AC generators; Oil refining; Coal extraction.
- *Inference.* **What Japan should export to the US:** Accumulator parts & separators; Rare-earth permanent magnets; Wafers; Battery materials (foil, electrolyte); Manganese dioxide (EMD); Piezo-electric quartz; Gas assembly; Electrolysers; Refrigeration & heat-pump compressors; Furnace parts; Oil assembly (ac generators); Coal components; Coal assembly.
- *Inference.* **Where tariffs would undermine allied specialisation:** no node where the buyer's protection priority is in its top third and the partner is the trusted supplier.
- *Inference.* **Procurement, standards and content rules to recognise mutually:** Electric Vehicles|Extraction, wind-turbine-generating-sets, accumulator-parts-and-separators, rare-earth-permanent-magnets, Oil|Extraction.
- *Inference.* **Where third-country circumvention safeguards are genuinely required:** Electric Vehicles|Refining, rare-earth-oxides-and-compounds; elsewhere they would add cost without security.
- *Inference.* **Where unrestricted global sourcing is acceptable:** uranium-fuel-upstream--aggregated, pgms-iridium-ruthenium, nuclear-special-materials, transformers, uranium-conversion-and-enrichment, copper-ores-and-concentrates.

Table A3 in the appendix (22 rows) carries this table in full; the CSV outputs carry every column.

- The tariff baseline is unsettled, which is itself a reason to bind the nodes above:
- EO signed 4 Sep 2025 sets 15% all-in tariff on Japanese goods (reciprocal) and a Section 232 auto/auto-parts rate that tops MFN up to 15% (added rate zero at or above 15%).[7]
- On 20 Feb 2026 the Court ruled 6-3 that IEEPA does not authorise tariffs, removing the legal basis of the 15% Japan reciprocal rate; a temporary Section 122 global surcharge replaced it; Section 232 duties (autos etc.)

unaffected.[11]

- Jiji reports the US imposed new 12.5% Section 301 tariffs on Japan and dozens of partners after the Section 122 surcharge expired 24 Jul 2026; Japan called it regrettable and the 15% auto rate is said to be maintained; a separate excess-capacity 301 probe (launched Mar 2026) includes Japan. Check USTR Federal Register notice for final rate.[13]

5. Investment Agenda

US → Japan investment (US capital and firms expanding in Japan)

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I09	Polysilicon	USA	JPN	JPN takes 25% of allied demand (15% of allied net imports); AIB class LOW INTERVENTION		strategic imports	medium	2-3 years
I10	Batteries components (accumulator parts & separators)	USA	JPN	supply-chain need 0.06 (USA) / 0.75 (JPN)		export credit	medium	2-3 years
I11	Wind assembly (electricity meters, overhead conductors, power cables)	USA	JPN	supply-chain need 0.33 (USA) / 0.75 (JPN)	Hitachi Energy; Mitsubishi Electric Power Products	export credit	low	2-3 years
I12	Batteries assembly (li-ion cells & packs, other accumulators)	USA	JPN	supply-chain need 0.90 (USA) / 0.71 (JPN)	Panasonic Energy; Toyota; Honda	strategic imports	low	2-3 years
I13	Refrigerants	USA	JPN	JPN takes 4% of allied demand (2% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I14	Sensors & controller parts	USA	JPN	JPN takes 12% of allied demand (1% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years

Sources: 1.

Japan → US investment (Japan capital and firms expanding in US)

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I01	Accumulator parts & separators	JPN	USA	USA takes 42% of allied demand (39% of allied net imports); AIB class BUILD		anchor offtake	medium	2-3 years
I02	Oil extraction	JPN	USA	supply-chain need 0.47 (USA) / 0.98 (JPN)		anchor offtake	high	2-3 years
I03	Wafers	JPN	USA	USA takes 13% of allied demand (0% of allied net imports); AIB class BUILD		standards alignment	high	2-3 years
I04	Battery materials (foil, electrolyte)	JPN	USA	USA takes 20% of allied demand (15% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I05	Manganese dioxide (EMD)	JPN	USA	USA takes 6% of allied demand (2% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I06	Electrolysers	JPN	USA	USA takes 13% of allied demand (3% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I07	Refrigeration & heat-pump compressors	JPN	USA	USA takes 22% of allied demand (12% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I08	Furnace parts	JPN	USA	USA takes 38% of allied demand (29% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years

Sources: 1.

Joint investment (at home or in third countries)

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	NODE	CAPITAL / CAPABILITY OWNER	DESTINATION	DEMAND CASE	PRIVATE ACTORS (EVIDENCE)	PUBLIC INSTRUMENT	ADDITIONALITY	HORIZON
I15	Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite)	USA+JPN	third countries (CHL; ARG; BRA) and either partner	supply-chain need 0.82 (USA) / 0.98 (JPN)		anchor offtake	high	2-3 years
I16	Wind turbine generating sets	USA+JPN	third countries (ESP; DNK) and either partner	JPN takes 2% of allied demand (2% of allied net imports); AIB class BUILD		anchor offtake	high	2-3 years
I17	Rare-earth permanent magnets	USA+JPN	third countries (PHL; VNM) and either partner	USA takes 10% of allied demand (9% of allied net imports); AIB class BUILD	Shin-Etsu; TDK; Proterial	anchor offtake	high	2-3 years
I18	Piezo-electric quartz	USA+JPN	third countries (TWN; PHL) and either partner	USA takes 16% of allied demand (11% of allied net imports); AIB class BUILD	Alcoa; Sojitz	anchor offtake	low	2-3 years

Sources: 1.

- *Inference.* Additionality = mean of three named components: the market failure (chokepoint 1, capacity constraint 0.75, gap or emerging bottleneck 0.5, none 0), one minus an existing commitment in the evidence layer (committed 1, announced 0.5) and one minus a working market (the buyer already takes 30% or more of its imports from the supplier). It is **low** whenever there is no failure or the investment is already committed: public support would then not be additional.

6. Third-Country and Development Cooperation

- From the development-partnership pairs (scripts/93: a donor's need meets a developing economy's credible Catch Up pathway in the same step), US and Japan are both donors at 10 recipient steps. Ranked: both donors' own need first, then steps inside a recommended compact.[1]
- *Inference.* Roles: US brings demand and offtake and coordination across compacts; Japan brings capital and export finance and specialised technology and components (AIB strategic roles).

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	RECIPIENT	TECHNOLOGY	STAGE	COMPACT	RECIPIENT CHOICE	RECIPIENT ANCHOR	US INTEREST	JAPAN INTEREST
X01	Malaysia	Electric Vehicles	Refining	SC-MAGNETS	catch up	direct supply share; direct export share	own secure need (USA)	own secure need (JPN)
X02	Thailand	Electric Vehicles	Refining	SC-MAGNETS	catch up	direct export share	own secure need (USA)	own secure need (JPN)

ID	RECIPIENT	TECHNOLOGY	STAGE	COMPACT	RECIPIENT CHOICE	RECIPIENT ANCHOR	US INTEREST	JAPAN INTEREST
X03	India	Wind	Refining	SC-WIND	catch up	direct export share	own partner need (USA)	own partner need (JPN)
X04	Indonesia	Batteries	Refining	SC-BATTERIES	preserve	direct supply share; direct export share; operating capacity; committed pipeline	allied-system need only	own catch up need (JPN)
X05	Argentina	Batteries	Refining	SC-BATTERIES	catch up	direct supply share; operating capacity; committed pipeline	allied-system need only	own catch up need (JPN)
X06	Democratic Republic of Congo	Batteries	Refining	SC-BATTERIES	catch up	direct export share; operating capacity	allied-system need only	own catch up need (JPN)
X07	Madagascar	Batteries	Refining	SC-BATTERIES	catch up	direct export share; operating capacity	allied-system need only	own catch up need (JPN)
X08	Malaysia	Batteries	Refining	SC-BATTERIES	catch up	direct export share	allied-system need only	own catch up need (JPN)
X09	Philippines	Batteries	Refining	SC-BATTERIES	catch up	direct export share; operating capacity	allied-system need only	own catch up need (JPN)
X10	India	Batteries	Refining	SC-BATTERIES	catch up	direct export share	allied-system need only	own catch up need (JPN)

Sources: 1.

- The trilateral Wagerup gallium project reportedly involves Sojitz and the Japanese government with an estimated $174M US Dept of War equity investment and groundbreaking on 2026-08-24; the equity announcement year is unclear.[48]
- AESC paused its $1.6B Florence County plant in June 2025 citing policy and market uncertainty.[40]
- US government will arrange financing and permitting for at least $80bn of new Westinghouse reactors (mainly AP1000) and receives a participation interest of 20% of distributions above $17.5bn convertible to up to 20% equity; date 28 Oct 2025 per most sources (one says 27 Oct).[19]
- LGES agreed to sell the Ohio JV factory building and related assets to Honda for about $2.85B with the JV continuing via lease and closing targeted for 2026-02-28.[45]
- Quad foreign ministers adopted a framework to mobilise up to $20bn for secure critical-mineral supply chains; a separate July 2025 Quad initiative is mentioned only in passing and not sourced.[3]

7. Recommended Sectoral Compacts

Inference. 5 compacts pass (rule: at least one joint bet, at least two active nodes and both countries holding an AIB compact role; or, labelled **evidence-led**, an existing bilateral pipeline or at least 2 bilateral project items where OSI measures a need but cannot see the partner's capability). The hypothesis test for every AIB compact:

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

RANK	COMPACT	VERDICT	JOINT BETS	COORDINATE	US ROLE	JAPAN ROLE	BILATERAL MECHANISMS	RECOMMENDED
1	Battery Materials & Cells	bilateral priority	2	3	Resource supplier	Processing anchor	USJ-CMF; USJ-CMAP; USJ-CMA; USJ-QUAD	TRUE
2	Permanent Magnets & Drive Systems	bilateral priority	2	1	Scale manufacturer	Scale manufacturer	USJ-CMF; USJ-CMAP; USJ-QUAD	TRUE
3	Solar Manufacturing	bilateral priority	1	1	Demand anchor	Specialized technology supplier		TRUE
4	Nuclear Fuel & Reactor Systems	evidence-led candidate	0	0	Specialized technology supplier	Capacity-building host	USJ-SII	TRUE
5	Grid Equipment & Conductors	evidence-led candidate	0	0	Demand anchor	Specialized technology supplier		TRUE
6	Wind & Hydropower Equipment	secondary	1	0	Demand anchor	Bridge / coordination partner		FALSE
7	Semiconductors & Power Electronics	secondary	0	2	Specialized technology supplier	Specialized technology supplier	USJ-CMF; USJ-CMAP	FALSE
8	Heat Pumps & Cooling	secondary	0	2	Processing anchor	Specialized technology supplier		FALSE
9	Industrial Electrification & Hydrogen	secondary	0	2	Demand anchor	Scale manufacturer		FALSE
10	Electric Vehicles	not a bilateral priority	0	0	Scale manufacturer	Scale manufacturer		FALSE

Sources: 1.

1. Battery Materials & Cells

Strategic objective

- AIB mission: Build missing capacity in Li-ion cells & packs and lithium chemicals and manganese dioxide (EMD) (refined chemicals); targeted: preserve and second-source specialized production in lithium chemicals (refined chemicals)..[1]
- *Recommendation.* Change the outcome at natural graphite, accumulator parts & separators, battery materials (foil, electrolyte): geopolitical chokepoint; trusted partner capacity constraint.

Why this pair?

- *Inference.* Natural graphite: Shared vulnerability (both need the node and neither is a strong supplier; CHN holds 82% of world supply). Accumulator parts & separators: Capacity-constrained trusted supply (JPN is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Battery materials (foil, electrolyte): Capacity-constrained trusted supply (JPN is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Manganese dioxide (EMD): Capacity-constrained

trusted supply (JPN is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). Battery recycling feed: Complementary specialisation (USA is strong here and the other country is strong at another step of the same chain).

- AIB compact roles: US Resource supplier (lead node battery-recycling-feed); Japan Processing anchor (lead node manganese-dioxide-emd).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	US POSITION	JAPAN POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	natural-graphite	AIB node	moderate	moderate	shared vulnerability	BUILD	anchor offtake
Refining	battery-materials-foil-electrolyte	AIB node	moderate	strong	capacity constrained trusted supply	BUILD	anchor offtake
Components	accumulator-parts-and-separators	AIB node	moderate	strong	capacity constrained trusted supply	BUILD	anchor offtake
Assembly	li-ion-cells-and-packs	AIB node	moderate	moderate	low strategic case	LOW INTERVENTION	
Deployment	Batteries/Deployment	OSI cell (no AIB node at this stage)	strong	strong	deployment demand	DEMAND CONTEXT	
Recovery	battery-recycling-feed	AIB node	strong	moderate	complementary specialization	COORDINATE	standards alignment

Sources: 1.

Reciprocal bargain

- US provides: supply of battery recycling feed; long-term demand and offtake for accumulator parts & separators, battery materials (foil, electrolyte), manganese dioxide (emd); procurement recognition of the partner as trusted supply.
- Japan provides: capacity and technology in accumulator parts & separators, battery materials (foil, electrolyte), manganese dioxide (emd); long-term demand for battery recycling feed; JBIC/NEXI finance for capacity in the partner market.
- Jointly: pooled offtake and finance for non-coercive natural graphite.
- Third-country partners needed: MDG, MOZ, DEU, KOR, MYS, TWN, GRC, ESP, HUN, POL.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): MDG at natural graphite; DEU at synthetic graphite (anode-grade); KOR at cathode materials & precursors; POL at li-ion cells & packs; FIN at cobalt intermediates & chemicals; CHL at lithium chemicals; VNM at other accumulators. The bilateral compact should not try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* anchor_offtake for natural graphite; anchor_offtake for accumulator parts & separators; anchor_offtake for battery materials (foil, electrolyte); anchor_offtake for manganese dioxide (emd); standards_alignment for battery recycling feed

Priority transactions

- *Recommendation.* Long-term offtake by Japan buyers for US electric vehicles extraction (cobalt ores, lithium minerals, natural graphite), sized to underwrite new capacity (candidate hosts: CHL, ARG, BRA) (D01).

Success by 2030

- Natural graphite: coercive share of supply to the two buyers below 50% (CHN now holds 82% of world supply).
- Accumulator parts & separators: allied residual gap under the China shock closed (now 37% of allied demand); Japan's share of US's imports up from 20%.
- Battery materials (foil, electrolyte): allied residual gap under the China shock closed (now 14% of allied demand); Japan's share of US's imports up from 7%.
- Manganese dioxide (EMD): allied residual gap under the China shock closed (now 34% of allied demand); Japan's share of US's imports up from 7%.
- Battery recycling feed: allied residual gap under the China shock closed (now 8% of allied demand); US's share of Japan's imports up from 23%.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- Signed 28 Mar 2023 and in force on signature without congressional approval; covers cobalt/graphite/lithium/manganese/nickel; no export duties; consult on non-market measures and forced-labour imports; Treasury/USTR treated it as an FTA for the IRA EV credit critical-minerals test.[5]
- AESC paused its $1.6B Florence County plant in June 2025 citing policy and market uncertainty.[40]
- Panasonic opened the De Soto plant in July 2025 with about 1100 employees and one of eight lines running versus a 4000-job target.[42]
- Toyota began production at its Liberty NC battery plant on 2025-11-12 (reported investment $13.9B with a further $10B pledged); output is mainly hybrid batteries and sources differ on line counts.[44]
- LGES agreed to sell the Ohio JV factory building and related assets to Honda for about $2.85B with the JV continuing via lease and closing targeted for 2026-02-28.[45]
- Panasonic plans to repurpose some US lines including De Soto to data-center energy storage with first cells expected mid-to-late 2028 or early 2029 while EV cell output continues; ramp is slower than planned.[46]

2. Permanent Magnets & Drive Systems

Strategic objective

- AIB mission: Build missing capacity in rare-earth permanent magnets (magnets & core steel)..[1]
- *Recommendation.* Change the outcome at rare-earth oxides & compounds, rare-earth permanent magnets, ac generators: geopolitical chokepoint; strategic manufacturing gap.

Why this pair?

- *Inference.* Rare-earth oxides & compounds: Shared vulnerability (both need the node and neither is a strong supplier; CHN holds 84% of world supply). Rare-earth permanent magnets: Capacity-constrained trusted supply (JPN is a capable trusted supplier, the other side needs the node, and allied capacity is short of demand). AC generators: Complementary specialisation (USA is strong here and the other country is strong at another step of the same chain).

- AIB compact roles: US Scale manufacturer (lead node ac-generators); Japan Scale manufacturer (lead node non-oriented-electrical-steel-ngo).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	US POSITION	JAPAN POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	rare-earth-ores-monazite	AIB node	moderate	weak	low strategic case	LOW INTERVENTION	
Refining	rare-earth-oxides-and-compounds	AIB node	moderate	moderate	shared vulnerability	BUILD	anchor offtake
Components	rare-earth-permanent-magnets	AIB node	moderate	strong	capacity constrained trusted supply	BUILD	anchor offtake
Assembly	ac-generators	AIB node	strong	weak	complementary specialization	COORDINATE	standards alignment
Deployment	Electric Vehicles/Deployment	OSI cell (no AIB node at this stage)	strong	strong	low strategic case	LOW INTERVENTION	
Recovery	Electric Vehicles/Recovery	OSI cell (no AIB node at this stage)	not measured	not measured	not assessed	INSUFFICIENT DATA	

Sources: 1.

Reciprocal bargain

- US provides: supply of ac generators; long-term demand and offtake for rare-earth permanent magnets; procurement recognition of the partner as trusted supply.
- Japan provides: capacity and technology in rare-earth permanent magnets; long-term demand for ac generators; JBIC/NEXI finance for capacity in the partner market.
- Jointly: pooled offtake and finance for non-coercive rare-earth oxides & compounds.
- Third-country partners needed: MYS, PHL, VNM.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): MYS at rare-earth oxides & compounds; VNM at rare-earth metals; DEU at ferrite & other magnets; DEU at electric motors; DEU at motor & generator parts; DEU at non-oriented electrical steel (ngo). The bilateral compact should not try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* anchor_offtake for rare-earth oxides & compounds; anchor_offtake for rare-earth permanent magnets; standards_alignment for ac generators

Priority transactions

- *Recommendation.* Long-term offtake by US and Japan buyers for non-coercive rare-earth oxides & compounds, sized to underwrite new capacity (candidate hosts: MYS) (D02).

Success by 2030

- Rare-earth oxides & compounds: coercive share of supply to the two buyers below 50% (CHN now holds 84% of world supply).
- Rare-earth permanent magnets: coercive share of supply to the two buyers below 50% (CHN now holds 64% of world supply); Japan's share of US's imports up from 8%.
- AC generators: allied residual gap under the China shock closed (now 4% of allied demand); US's share of Japan's imports up from 24%.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- Framework text dated 27 Oct 2025 (signed Tokyo 28 Oct) commits to financing within six months for selected projects (grants/guarantees/loans/equity/offtake/insurance), a ministerial within 180 days, faster permitting, pricing measures to support non-China supply, scrap recycling, a complementary stockpile arrangement, and a Rapid Response Group led by the US Energy Secretary and METI Minister; names no dollar amounts and is non-binding.[16]
- Trump and Takaichi signed a framework on 2025-10-28 in Tokyo covering permitting streamlining, coordinated investment and a Supply Security Rapid Response Group led by the US Energy Secretary and Japan's METI minister.[49]

3. Solar Manufacturing

Strategic objective

- AIB mission: Build missing capacity in modules, cells & cover glass and polysilicon..[1]
- *Recommendation.* Change the outcome at polysilicon, wafers: geopolitical chokepoint.

Why this pair?

- *Inference.* Polysilicon: US supplies / Japan needs (USA is a capable trusted supplier and the other side needs the node). Wafers: Complementary specialisation (JPN is strong here and the other country is strong at another step of the same chain).
- AIB compact roles: US Demand anchor (lead node solar-cells); Japan Specialized technology supplier (lead node silver-powder).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	US POSITION	JAPAN POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	Solar/Extraction	OSI cell (no AIB node at this stage)	moderate	weak	low strategic case	LOW INTERVENTION	
Refining	polysilicon	AIB node	strong	moderate	a supplies b needs	COORDINATE	strategic imports
Components	solar-cells	AIB node	moderate	moderate	low strategic case	LOW INTERVENTION	

STAGE	UNIT	BASIS	US POSITION	JAPAN POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Assembly	pv-modules-and-generators	AIB node	moderate	weak	low strategic case	LOW INTERVENTION	
Deployment	Solar/Deployment	OSI cell (no AIB node at this stage)	strong	strong	deployment demand	DEMAND CONTEXT	
Recovery	Solar/Recovery	OSI cell (no AIB node at this stage)	not measured	not measured	not assessed	INSUFFICIENT DATA	

Sources: 1.

Reciprocal bargain

- US provides: supply of polysilicon; long-term demand and offtake for wafers; procurement recognition of the partner as trusted supply.
- Japan provides: capacity and technology in wafers; long-term demand for polysilicon; JBIC/NEXI finance for capacity in the partner market.
- Jointly: common specifications and qualification.
- Third-country partners needed: MYS, DEU, TWN.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): THA at solar cells; VNM at pv modules & generators; MYS at pv cover glass. The bilateral compact should not try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* strategic_imports for polysilicon; standards_alignment for wafers

Priority transactions

- *Recommendation.* Multi-year supply agreements for US polysilicon into Japan buyers (D05).
- *Recommendation.* Joint specification work plan for wafers (D10).

Success by 2030

- Polysilicon: coercive share of supply to the two buyers below 50% (CHN now holds 97% of world supply); US's share of Japan's imports up from 57%.
- Wafers: coercive share of supply to the two buyers below 50% (CHN now holds 98% of world supply); Japan's share of US's imports up from 7%.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

4. Nuclear Fuel & Reactor Systems (evidence-led)

- *Inference.* **Basis: external evidence.** OSI trade data show no bilateral complementarity at this compact's nodes (Japan world share: uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores) 0.0%, uranium conversion & enrichment 0.1%, nuclear reactors & enrichment machinery

0.0%, fuel assemblies & zirconium 0.7%, reactor parts 0.0%, nuclear special materials 4.8%); the case rests on the bilateral pipeline and corporate evidence listed at the end of this compact.

- Need that OSI does measure in the compact: uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores) (US high 0.99, 91% of allied demand; Japan low 0.61); uranium conversion & enrichment (US high 0.99, 42% of allied demand; Japan low (cell 0.97; not a material buyer)); nuclear reactors & enrichment machinery (US demand 0.47, 8% of allied demand; Japan low 0.66); fuel assemblies & zirconium (US low 0.48, 7% of allied demand; Japan demand 0.61); nuclear special materials (US demand n/a, 51% of allied demand; Japan low n/a).[1]

Strategic objective

- AIB mission: Build missing capacity in reactors & enrichment machinery and nuclear special materials (fuel & nuclear materials); coordinate trade to redirect existing supply in reactor components; preserve and second-source specialized production in reactors & enrichment machinery..[1]
- *Recommendation.* Use USJ-SII to tie the announced projects to the nodes where OSI measures need, and verify the partner capability that trade data cannot see (ownership abroad, domestic sales, heavy components) before any public finance is added.

Why this pair?

- *Inference.* No OSI-observed complementarity; see the evidence below.
- AIB compact roles: US Specialized technology supplier (lead node reactor-parts); Japan Capacity-building host (lead node NA).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	US POSITION	JAPAN POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	uranium-fuel-upstream--aggregated	AIB node	weak	weak	low strategic case	LOW INTERVENTION	
Refining	uranium-conversion-and-enrichment	AIB node	moderate	weak	low strategic case	LOW INTERVENTION	
Components	fuel-assemblies-and-zirconium	AIB node	moderate	weak	low strategic case	LOW INTERVENTION	
Assembly	nuclear-reactors-and-enrichment-machinery	AIB node	weak	weak	low strategic case	LOW INTERVENTION	
Deployment	Nuclear/Deployment	OSI cell (no AIB node at this stage)	strong	strong	deployment demand	DEMAND CONTEXT	
Recovery	Nuclear/Recovery	OSI cell (no AIB node at this stage)	not measured	not measured	not assessed	INSUFFICIENT DATA	

Sources: 1.

Reciprocal bargain

- US provides: procurement recognition of the partner as trusted supply.
- Japan provides: JBIC/NEXI finance for capacity in the partner market.

- Jointly: common specifications and qualification.
- Third-country partners needed: none identified by the AIB host model.

Policy instruments (minimum necessary)

- *Recommendation.* Use the existing investment vehicle; no new instrument.

Priority transactions

- See Section 9; no OSI-ranked deal in this compact.

Success by 2030

- Projects in the pipeline reach final investment decision with partner-made equipment in the bill of materials.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- US government will arrange financing and permitting for at least $80bn of new Westinghouse reactors (mainly AP1000) and receives a participation interest of 20% of distributions above $17.5bn convertible to up to 20% equity; date 28 Oct 2025 per most sources (one says 27 Oct).[19]
- Announced 28 Oct 2025 as preliminary pledges in a launch phase needing review; AA lists nuclear up to $332bn, Mitsubishi Electric AI infrastructure $30bn and Panasonic energy storage $15bn.[20]
- The joint fact sheet lists Japanese-linked interest of up to about $490B in 'launch phase' projects, with Lutnick saying projects still need approval and final selection rests with the US.[50]
- The fact sheet lists up to $100B for Westinghouse AP1000/SMR with possible Japanese supplier involvement (MHI Toshiba IHI); CSIS cites a separate figure of $80B in a US-Westinghouse partnership.[21]
- The fact sheet lists up to $100B for GE Vernova Hitachi BWRX-300 SMR construction with Hitachi GE Vernova as a possible Japanese participant.[22]
- ANS (17 Mar 2026) says the 2025 US-Japan agreement projected up to $100bn for Westinghouse reactors but specific projects were not yet identified and Westinghouse is still working transactions.[28]
- Second batch announced 19 Mar 2026 at the Takaichi-Trump summit lists SMRs by GE Vernova Hitachi in Tennessee and Alabama at up to $40bn; TVA Clinch River and Bellefonte/Widows Creek siting is from secondary reports.[29]

5. Grid Equipment & Conductors (evidence-led)

- *Inference.* **Basis: external evidence.** OSI trade data show no bilateral complementarity at this compact's nodes (Japan world share: copper smelting & refining 6.2%, grain-oriented electrical steel (goes) 25.1%, insulators & bushings 4.1%, electricity meters 0.0%, transformers 0.0%, switchgear & protection 1.8%, copper ores & concentrates 0.0%, copper wire & rod 1.1%, overhead conductors 0.0%, power cables 1.8%, aluminium wire rod 0.0%); the case rests on the bilateral pipeline and corporate evidence listed at the end of this compact.
- Need that OSI does measure in the compact: copper smelting & refining (US high 0.90, 15% of allied demand; Japan low (cell 0.71; not a material buyer)); insulators & bushings (US demand 0.32, 19% of allied demand; Japan low 0.60); electricity meters (US demand 0.33, 22% of allied demand; Japan low (cell 0.75; not a material buyer)); transformers (US demand 0.33, 40% of allied demand; Japan low (cell 0.75; not a material buyer)); switchgear & protection (US demand 0.33, 20% of allied demand; Japan low (cell 0.75; not a material buyer));

copper ores & concentrates (US low (cell 0.89; not a material buyer), 0% of allied demand; Japan high 0.98); copper wire & rod (US high 0.96, 10% of allied demand; Japan low (cell 0.95; not a material buyer)); overhead conductors (US demand 0.33, 32% of allied demand; Japan low (cell 0.75; not a material buyer)); power cables (US demand 0.33, 20% of allied demand; Japan low (cell 0.75; not a material buyer)); aluminium wire rod (US demand n/a, 24% of allied demand; Japan not_measured n/a).[1]

Strategic objective

- AIB mission: Monitor: no intervention class reaches the materiality floor (25% of the compact's supply to shift or US$1bn); targeted: build missing capacity in grain-oriented electrical steel (GOES) (wire rod & transformer steel)..[1]
- *Recommendation.* Use the existing bilateral channels to tie the announced projects to the nodes where OSI measures need, and verify the partner capability that trade data cannot see (ownership abroad, domestic sales, heavy components) before any public finance is added.

Why this pair?

- *Inference.* No OSI-observed complementarity; see the evidence below.
- AIB compact roles: US Demand anchor (lead node transformers); Japan Specialized technology supplier (lead node grain-oriented-electrical-steel-goes).[1]

Supply-chain architecture

STAGE	UNIT	BASIS	US POSITION	JAPAN POSITION	DESIRED BILATERAL ARRANGEMENT	TREATMENT	INSTRUMENT
Extraction	copper-ores-and-concentrates	AIB node	moderate	weak	low strategic case	LOW INTERVENTION	
Refining	copper-smelting-and-refining	AIB node	weak	moderate	low strategic case	LOW INTERVENTION	
Components	insulators-and-bushings	AIB node	moderate	moderate	low strategic case	LOW INTERVENTION	
Assembly	electricity-meters	AIB node	moderate	weak	low strategic case	LOW INTERVENTION	
Deployment	Electric Grid/Deployment	OSI cell (no AIB node at this stage)	not measured	not measured	low strategic case	LOW INTERVENTION	
Recovery	Electric Grid/Recovery	OSI cell (no AIB node at this stage)	not measured	not measured	not assessed	INSUFFICIENT DATA	

Sources: 1.

Reciprocal bargain

- US provides: procurement recognition of the partner as trusted supply.
- Japan provides: JBIC/NEXI finance for capacity in the partner market.
- Jointly: common specifications and qualification.
- Third-country partners needed: none identified by the AIB host model.
- Better placed than this pair (largest trusted supplier once China is removed, AIB): ITA at insulators & bushings; MEX at electricity meters. The bilateral compact should not try to replace them there.[1]

Policy instruments (minimum necessary)

- *Recommendation.* Use the existing investment vehicle; no new instrument.

Priority transactions

- *Recommendation.* Long-term offtake by US and Japan buyers for non-coercive electric grid refining (copper smelting & refining, copper wire & rod, grain-oriented electrical steel (goes)), sized to underwrite new capacity (candidate hosts: DEU, KOR, TWN) (D07).

Success by 2030

- Projects in the pipeline reach final investment decision with partner-made equipment in the bill of materials.
- The AIB node class moves from BUILD to COORDINATE or LOW INTERVENTION on the next run.

Failure / exit conditions

- No signed offtake or financing within 24 months of launch: drop the node from the compact.
- The coercive share of world supply falls below the chokepoint line, or the AIB class turns LOW INTERVENTION: move to open-market treatment.
- Projects proceed on commercial terms without public support: stop the support (it is not additional).

Evidence (external)

- Hitachi Energy pledged more than $1B for US grid equipment manufacturing including the $457M South Boston plant.[36]
- Hitachi Energy is reported to commit $528M to a Gallman Mississippi power transformer plant supporting 650+ jobs, with no capacity or timeline released.[37]
- Mitsubishi Electric opened a roughly $200M Memphis Tennessee transformer factory in 2013.[38]
- Mitsubishi Electric announced an $86M advanced switchgear factory in Western Pennsylvania in October 2024.[39]
- Nippon Steel committed about $11B of new US Steel investment by 2028 (some reports cite $14B) with electrical steel mentioned mainly for automotive.[41]
- Nikkei reported in November 2025 that US Steel would build GOES capability at Big River Osceola Arkansas with mass production in 2028; US Steel confirmed the project to Argus without investment or capacity details (date approximate).[43]
- Hitachi Energy broke ground on 2026-06-29 on a $457M South Boston expansion creating about 825 jobs; the site already employs about 850 and completion date was not stated on the page (other reports say 2028).[47]

8. Master Energy Agreement Assessment

- *Inference.* Rule: **existing** if every recommended compact is covered by a mechanism with both a pipeline and finance and a cross-sector mechanism has KPIs and a review cycle; **upgrade** if at least 50% of compacts are covered by a mechanism with finance or a pipeline, or a cross-sector mechanism with a pipeline exists; **new MEA** otherwise.
- *Inference.* Result: **upgrade**; coverage 60% of recommended compacts; cross-sector mechanism with KPIs and review: no.

COMPACT	ANY MECHANISM	FINANCE OR PIPELINE	PIPELINE AND FINANCE
SC-BATTERIES	TRUE	TRUE	FALSE
SC-MAGNETS	TRUE	TRUE	FALSE
SC-SOLAR	FALSE	FALSE	FALSE
SC-NUCLEAR	TRUE	TRUE	TRUE

COMPACT	ANY MECHANISM	FINANCE OR PIPELINE	PIPELINE AND FINANCE
SC-GRID	FALSE	FALSE	FALSE

Sources: 1.

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

ID	MECHANISM	STATUS	SCOPE	PIPELINE	FINANCE	KPIS	REVIEW	RECIPROCAL
USJ-SII	Japan-US Strategic Investment Initiative (MOU of 4 Sep 2025)	in force	SC-NUCLEAR/cross sector	TRUE	TRUE	FALSE	partial	FALSE
USJ-CMF	US-Japan critical minerals and rare earths framework (Oct 2025) with the Supply Security Rapid Response Group	signed	SC-MAGNETS/SC-BATTERIES/SC-SEMIS	partial	TRUE	FALSE	TRUE	TRUE
USJ-CMAP	US-Japan Action Plan on Critical Minerals (19 Mar 2026)	signed	SC-MAGNETS/SC-BATTERIES/SC-SEMIS	FALSE	FALSE	FALSE	partial	TRUE
USJ-CMA	US-Japan Critical Minerals Agreement (28 Mar 2023)	in force	SC-BATTERIES	FALSE	FALSE	FALSE	FALSE	TRUE
USJ-TRADE	US-Japan trade agreements (USJTA and DTA 2019; July 2025 deal and EO 14345)	in force	cross sector	FALSE	FALSE	FALSE	FALSE	partial
USJ-EPC	US-Japan Economic Policy Consultative Committee (Economic 2+2)	in force	cross sector	FALSE	FALSE	FALSE	partial	TRUE
USJ-DFCJBIC	DFC-JBIC cooperation memorandum	in force	cross sector	FALSE	TRUE	FALSE	FALSE	TRUE
USJ-QUAD	Quad Critical Minerals Initiative Framework (May 2026)	announced	SC-MAGNETS/SC-BATTERIES	FALSE	TRUE	FALSE	FALSE	TRUE

Sources: 1.

Recommended: upgrade Japan-US Strategic Investment Initiative (MOU of 4 Sep 2025) into the execution framework for the compacts

- **Scope:** Battery Materials & Cells; Permanent Magnets & Drive Systems; Solar Manufacturing; Nuclear Fuel & Reactor Systems; Grid Equipment & Conductors; no other sectors.
- **Reciprocal commitments:** US commits demand certainty (procurement recognition and anchor offtake for the nodes where Japan supplies) and market-access predictability on those nodes; Japan commits capacity and JBIC/NEXI/JOGMEC finance for the same nodes, in the US or with named third-country hosts, and offtake for US-supplied nodes.
- **Compact portfolio:** Battery Materials & Cells (bilateral priority); Permanent Magnets & Drive Systems (bilateral priority); Solar Manufacturing (bilateral priority); Nuclear Fuel & Reactor Systems (evidence-led candidate); Grid Equipment & Conductors (evidence-led candidate).
- **Financing lanes:** project SPV loans and insurance as in the existing vehicle (JBIC, NEXI); US loan guarantees and offtake (DOE LPO, DoW OSC, DLA); third-country finance (DFC with JBIC).
- **Governance:** one bilateral steering line inside USJ-SII, reporting to the Economic 2+2; the critical-minerals Rapid Response Group runs the minerals and magnets workstreams. Selection by joint committee, not by one side (the 2026 selection dispute is evidence the current process is not reciprocal).

- **Milestones:** the 12-month milestones of the deal pipeline (Section 9).
- **KPI dashboard:** per compact node, the coercive share of each country's supply, the bilateral import share, offtake signed (volume), capacity at FID and operating, and public money committed vs private.
- **Review cycle:** every 12 months, with the OSI tables re-run on new trade data.
- **Sunset / exit:** each compact carries the exit conditions in Section 7; the framework expires with the investment vehicle (2029-01-19) unless renewed on evidence.

9. 12–36 Month Deal Pipeline

Inference. Ranked lexicographically on separate fields (importance, then additionality, urgency, tractability, execution speed, value-chain interdependency, evidence quality); urgency is not importance. At most two deals per compact.

Some columns of the next table are left out of the print edition; the Markdown and CSV outputs carry them all.

RANK	DEAL / ACTION	COMPACT	INSTRUMENT	RESPONSIBLE INSTITUTIONS	12M MILESTONE	IMPORTANCE	URGENCY	ADDITIONALITY
1	Long-term offtake by Japan buyers for US electric vehicles extraction (cobalt ores, lithium minerals, natural graphite), sized to underwrite new capacity (candidate hosts: CHL, ARG, BRA)	SC-BATTERIES	anchor offtake	USA: DOE, DoW OSC, DLA; JPN: METI, JOGMEC	Signed long-term offtake from at least one large buyer	high	high	high
2	Long-term offtake by US and Japan buyers for non-coercive rare-earth oxides & compounds, sized to underwrite new capacity (candidate hosts: MYS)	SC-MAGNETS	anchor offtake	USA: DOE, DoW OSC, DLA; JPN: METI, JOGMEC	Signed long-term offtake from at least one large buyer	high	high	high

RANK	DEAL / ACTION	COMPACT	INSTRUMENT	RESPONSIBLE INSTITUTIONS	12M MILESTONE	IMPORTANCE	URGENCY	ADDITIONALITY
3	Long-term offtake by Japan buyers for US wind turbine generating sets, sized to underwrite new capacity (candidate hosts: ESP, DNK)	SC-WIND	anchor offtake	USA: DOE, DoW OSC, DLA; JPN: METI, JOGMEC	Signed long-term offtake from at least one large buyer	high	high	high
4	Long-term offtake by Japan buyers for US oil extraction, sized to underwrite new capacity		anchor offtake	USA: DOE, DoW OSC, DLA; JPN: METI, JOGMEC	Signed long-term offtake from at least one large buyer	high	low	high
5	Multi-year supply agreements for US polysilicon into Japan buyers	SC-SOLAR	strategic imports	USA: USTR, Commerce; JPN: METI	Named import lanes and volumes agreed between buyers and the supplier's producers	high	high	medium
6	Long-term offtake by US buyers for Japan piezo-electric quartz, sized to underwrite new capacity (candidate hosts: TWN, PHL)	SC-SEMIS	anchor offtake	USA: DOE, DoW OSC, DLA; JPN: METI, JOGMEC	Signed long-term offtake from at least one large buyer	high	medium	low

RANK	DEAL / ACTION	COMPACT	INSTRUMENT	RESPONSIBLE INSTITUTIONS	12M MILESTONE	IMPORTANCE	URGENCY	ADDITIONALITY
7	Long-term offtake by US and Japan buyers for non-coercive electric grid refining (copper smelting & refining, copper wire & rod, grain-oriented electrical steel (goes)), sized to underwrite new capacity (candidate hosts: DEU, KOR, TWN)	SC-GRID	anchor offtake	USA: DOE, DoW OSC, DLA; JPN: METI, JOGMEC	Signed long-term offtake from at least one large buyer	high	medium	low
8	Long-term offtake by Japan buyers for US refrigerants, sized to underwrite new capacity (candidate hosts: NLD)	SC-HEAT	anchor offtake	USA: DOE, DoW OSC, DLA; JPN: METI, JOGMEC	Signed long-term offtake from at least one large buyer	medium	low	high
9	Long-term offtake by US buyers for Japan electrolysers, sized to underwrite new capacity (candidate hosts: DEU)	SC-INDUSTRY	anchor offtake	USA: DOE, DoW OSC, DLA; JPN: METI, JOGMEC	Signed long-term offtake from at least one large buyer	medium	low	high
10	Joint specification work plan for wafers	SC-SOLAR	standards alignment	USA: USTR, NRC; JPN: METI	Joint list of specifications that fragment the market and a work plan to align them	low	high	high

Sources: 1.

10. What Not to Do

- *Recommendation.* **N01** Do not co-subsidise new Automatic controllers capacity: both are strong suppliers (US 16%, Japan 11%). Trusted supply is adequate: this is commercial competition.

- *Recommendation.* **N02** Do not co-subsidise new Gas components capacity: both are strong suppliers (US 21%, Japan 4%). Trusted supply is adequate: this is commercial competition.
- *Recommendation.* **N03** Do not co-subsidise new Semiconductor manufacturing equipment capacity: both are strong suppliers (US 16%, Japan 21%). Trusted supply is adequate: this is commercial competition.
- *Recommendation.* **N04** Do not co-subsidise new Silver powder capacity: both are strong suppliers (US 26%, Japan 59%). Trusted supply is adequate: this is commercial competition.
- *Recommendation.* **N05** Do not create a new bilateral energy MOU or a free-standing MEA institution: USJ-SII already carries a pipeline and finance; dock the compacts in it.
- *Recommendation.* **N06** Do not put new public money behind Batteries assembly (li-ion cells & packs, other accumulators) investment that is already committed (evidence UJ-034; UJ-035; UJ-036; UJ-037; UJ-038); support is additional only beyond it.

11. Evidence Register

- Every recommendation (IDs J, C, T, I, X, D, N, M) has rows in evidence_register.csv: the OSI metrics behind it (variable, value for each country, year, source, evidence specificity) and any external items, with URL and access date. Recommendations that rest on external research only carry basis = external.[1]
- External evidence vintage: accessed 2026-11-06; 85 items (partial 56, unverified 13, verified 16).[1]

Where the findings are weak or counterintuitive

- Joint bets resting on a node need inherited from its cell (a proxy): Rare-earth oxides & compounds; Natural graphite; Polysilicon; Wind turbine generating sets.
- Evidence-led compacts (OSI cannot see the partner's capability in trade data): Nuclear Fuel & Reactor Systems; Grid Equipment & Conductors.
- Japan has no listed trade position (under 1% of world supply) at nodes where the US has material demand: solar cells, pv modules & generators, uranium/fuel extraction: aggregated low-coverage sub-sectors (spent fuel & nuclear scrap, uranium ores), uranium conversion & enrichment, nuclear reactors & enrichment machinery, lithium chemicals, pv cover glass, wind towers, heat pumps & ac units, solar thermal water heaters, electricity meters, transformers, hydro turbine parts, blade core wood (balsa), overhead conductors, aluminium wire rod, direct-reduced iron (dri/hbi). Trade data miss capability that is owned abroad or sold at home, so these nodes cannot show complementarity even where corporate evidence suggests it.
- OSI reads transformers as low strategic case, but the evidence layer records a capacity problem the annual trade data cannot see: Wood Mackenzie projected a 30% supply deficit for power transformers and 10% for distribution units in 2025 with imports about 80% of power transformer supply and Q2 2025 lead times of 128 weeks (GSUs 144 weeks). Treat it as an emerging bottleneck to re-test, not a settled open-market node.[51]
- OSI reads grain-oriented-electrical-steel-goes as low strategic case, but the evidence layer records a capacity problem the annual trade data cannot see: Cleveland-Cliffs states Butler Works is the only US producer of regular grain-oriented electrical steel used in power and distribution transformers. Treat it as an emerging bottleneck to re-test, not a settled open-market node.[52]
- Low-evidence units among the active states: none.
- Fuel-trade lanes reach tier 1 on Japanese import need (oil extraction): real energy-security lanes, but outside the electro-industrial compacts and with no market failure beyond the supply concentration the engine measures.

Sources

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Appendix — evidence tables and method

The tables below carry the full complementarity map, the node profiles and the trade actions that the main text summarises. Method, decision rules and limitations: docs/partnership_memos.md. Machine-readable outputs: country_profiles.csv, dyad_complementarity.csv, compact_candidates.csv, trade_priorities.csv, investment_priorities.csv, third_country_opportunities.csv, deal_pipeline.csv and evidence_register.csv.

Table A1. Section 2, What Each Country Is Good At — and What It Needs

COMPACT	NODE	STAGE	US SHARE	US NEED	JAPAN SHARE	JAPAN NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-BATTERIES	Natural graphite	Extraction	3.1%	high 0.82	1.7%	high 0.98	BUILD	CHN 82%
SC-BATTERIES	Battery recycling feed	Recovery	43.7%	low (cell 0.88; not a material buyer)	1.6%	low (cell 0.98; not a material buyer)	LOW INTERVENTION	CHN 8%
SC-BATTERIES	Lithium minerals	Extraction	4.0%	low (cell 0.82; not a material buyer)	0.4%	high 0.98	LOW INTERVENTION	CHN 25%
SC-BATTERIES	Manganese dioxide (EMD)	Refining	3.7%	demand 0.45	18.3%	low (cell 0.94; not a material buyer)	BUILD	CHN 37%
SC-BATTERIES	Cobalt intermediates & chemicals	Refining	2.4%	demand 0.45	2.4%	high 0.94	BUILD	CHN 76%
SC-BATTERIES	Lithium chemicals	Refining	3.1%	demand 0.45	0.1%	high 0.94	BUILD	CHN 74%
SC-BATTERIES	Nickel intermediates & chemicals	Refining	0.4%	low 0.45	2.9%	high 0.94	LOW INTERVENTION	CHN 37%
SC-BATTERIES	Accumulator parts & separators	Components	3.1%	high 0.73	11.0%	low (cell 0.75; not a material buyer)	BUILD	CHN 38%
SC-BATTERIES	Battery materials (foil, electrolyte)	Refining	2.8%	demand 0.45	11.2%	low (cell 0.94; not a material buyer)	BUILD	CHN 16%
SC-BATTERIES	Synthetic graphite (anode-grade)	Refining	5.7%	demand 0.45	6.7%	low (cell 0.94; not a material buyer)	BUILD	CHN 94%
SC-BATTERIES	Cathode materials & precursors	Refining	4.9%	low 0.45	9.7%	low (cell 0.94; not a material buyer)	BUILD	CHN 92%
SC-BATTERIES	Li-ion cells & packs	Assembly	6.4%	high 0.90	3.3%	low (cell 0.71; not a material buyer)	BUILD	CHN 84%
SC-BATTERIES	Other accumulators	Assembly	7.4%	high 0.90	7.6%	low (cell 0.71; not a material buyer)	BUILD	CHN 28%
SC-EV	Electric & plug-in cars	Assembly	7.1%	demand 0.15	7.3%	low 0.65	BUILD	CHN 69%
SC-EV	Electric two-wheelers & works trucks	Assembly	5.4%	demand 0.15	1.3%	low 0.65	BUILD	CHN 30%
SC-EV	Electric buses, trucks & tractors	Assembly	14.6%	low 0.15	0.0%	low 0.65	BUILD	CHN 24%
SC-EV	Electric rail vehicles	Assembly	0.5%	low n/a	3.1%	low n/a	LOW INTERVENTION	CHN 14%
SC-GRID	Copper ores & concentrates	Extraction	4.7%	low (cell 0.89; not a material buyer)	0.0%	high 0.98	LOW INTERVENTION	CHN 13%
SC-GRID	Copper smelting & refining	Refining	1.4%	high 0.90	6.2%	low (cell 0.71; not a material buyer)	LOW INTERVENTION	CHN 50%
SC-GRID	Grain-oriented electrical steel (GOES)	Refining	3.7%	low (cell 0.90; not a material buyer)	25.1%	low (cell 0.71; not a material buyer)	BUILD	CHN 41%
SC-GRID	Copper wire & rod	Refining	4.9%	high 0.96	1.1%	low (cell 0.95; not a material buyer)	LOW INTERVENTION	CHN 6%
SC-GRID	Aluminium wire rod	Refining	0.1%	demand n/a	0.0%	not measured n/a	LOW INTERVENTION	RUS 7%
SC-GRID	Insulators & bushings	Components	6.6%	demand 0.32	4.1%	low 0.60	BUILD	CHN 32%
SC-GRID	Electricity meters	Assembly	6.3%	demand 0.33	0.0%	low (cell 0.75; not a material buyer)	BUILD	CHN 34%

COMPACT	NODE	STAGE	US SHARE	US NEED	JAPAN SHARE	JAPAN NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-GRID	Transformers	Assembly	2.0%	demand 0.33	0.0%	low (cell 0.75; not a material buyer)	LOW INTERVENTION	CHN 21%
SC-GRID	Switchgear & protection	Assembly	5.4%	demand 0.33	1.8%	low (cell 0.75; not a material buyer)	LOW INTERVENTION	CHN 20%
SC-GRID	Overhead conductors	Components	1.0%	demand 0.33	0.0%	low (cell 0.75; not a material buyer)	LOW INTERVENTION	CHN 24%
SC-GRID	Power cables	Components	6.7%	demand 0.33	1.8%	low (cell 0.75; not a material buyer)	LOW INTERVENTION	CHN 19%
SC-HEAT	Refrigerants	Refining	19.9%	low n/a	1.4%	demand n/a	BUILD	CHN 45%
SC-HEAT	Refrigeration & heat-pump compressors	Components	7.6%	demand n/a	11.2%	low n/a	BUILD	CHN 33%
SC-HEAT	Heat pumps & AC units	Assembly	4.0%	demand n/a	0.9%	demand n/a	BUILD	CHN 49%
SC-HEAT	Solar thermal water heaters	Assembly	0.4%	demand n/a	0.0%	not measured n/a	BUILD	CHN 45%
SC-INDUSTRY	PGMs (iridium, ruthenium)	Refining	4.0%	demand n/a	1.1%	demand n/a	LOW INTERVENTION	RUS 30%
SC-INDUSTRY	Electrolysers	Assembly	7.5%	demand n/a	13.4%	low n/a	BUILD	CHN 32%
SC-INDUSTRY	Hydrogen, ammonia & methanol	Refining	8.7%	low n/a	0.0%	demand n/a	LOW INTERVENTION	RUS 3%
SC-INDUSTRY	Furnace parts	Components	7.7%	demand n/a	11.0%	low n/a	BUILD	CHN 21%
SC-INDUSTRY	Graphite electrodes	Components	2.3%	demand n/a	7.2%	low n/a	COORDINATE	CHN 38%
SC-INDUSTRY	Direct-reduced iron (DRI/HBI)	Refining	5.2%	demand n/a	0.0%	not measured n/a	LOW INTERVENTION	RUS 20%
SC-INDUSTRY	Electric furnaces & heating	Assembly	8.0%	demand n/a	5.5%	low n/a	BUILD	CHN 27%
SC-MAGNETS	Rare-earth ores (monazite)	Extraction	9.2%	low (cell 0.82; not a material buyer)	0.0%	high 0.98	INSUFFICIENT DATA	CHN 59%
SC-MAGNETS	Rare-earth oxides & compounds	Refining	5.3%	high 0.96	6.3%	high 1.00	BUILD	CHN 84%
SC-MAGNETS	Rare-earth metals	Refining	5.3%	low (cell 0.96; not a material buyer)	2.4%	high 1.00	BUILD	CHN 84%
SC-MAGNETS	Rare-earth permanent magnets	Components	1.8%	high 0.73	8.8%	demand 0.62	BUILD	CHN 64%
SC-MAGNETS	Ferrite & other magnets	Components	2.8%	high 0.73	5.4%	low 0.62	BUILD	CHN 58%
SC-MAGNETS	Non-oriented electrical steel (NGO)	Refining	1.4%	low (cell 0.90; not a material buyer)	9.2%	low (cell 0.71; not a material buyer)	BUILD	CHN 26%
SC-MAGNETS	Motor & generator parts	Components	6.5%	high 0.73	7.3%	low 0.62	BUILD	CHN 26%
SC-MAGNETS	AC generators	Assembly	27.4%	low 0.27	1.5%	low 0.60	LOW INTERVENTION	CHN 15%
SC-MAGNETS	Electric motors	Assembly	5.4%	high 0.73	5.5%	low 0.62	BUILD	CHN 26%
SC-NUCLEAR	Uranium/fuel Extraction: aggregated low-coverage sub-sectors (Spent fuel & nuclear scrap, Uranium ores)	Extraction	0.7%	high 0.99	0.0%	low 0.61	INSUFFICIENT DATA	RUS 8%
SC-NUCLEAR	Uranium conversion & enrichment	Refining	3.3%	high 0.99	0.1%	low (cell 0.97; not a material buyer)	LOW INTERVENTION	RUS 16%
SC-NUCLEAR	Fuel assemblies & zirconium	Components	9.2%	low 0.48	0.7%	demand 0.61	LOW INTERVENTION	RUS 28%
SC-NUCLEAR	Nuclear special materials	Refining	3.5%	demand n/a	4.8%	low n/a	LOW INTERVENTION	0%
SC-NUCLEAR	Reactor parts	Components	18.1%	low 0.48	0.0%	low 0.61	LOW INTERVENTION	RUS 45%
SC-NUCLEAR	Nuclear reactors & enrichment machinery	Assembly	0.7%	demand 0.47	0.0%	low 0.66	LOW INTERVENTION	RUS 49%
SC-SEMIS	Piezo-electric quartz	Refining	3.3%	demand n/a	10.8%	demand n/a	BUILD	CHN 60%
SC-SEMIS	Gallium, germanium & indium	Refining	6.5%	demand n/a	3.3%	demand n/a	BUILD	CHN 25%

COMPACT	NODE	STAGE	US SHARE	US NEED	JAPAN SHARE	JAPAN NEED	AIB CLASS	COERCIVE TOP SUPPLIER
SC-SEMIS	Semiconductor manufacturing equipment	Assembly	15.6%	low 0.20	21.2%	low 0.55	LOW INTERVENTION	CHN 5%
SC-SEMIS	Sensors & controller parts	Components	10.1%	low n/a	8.6%	demand n/a	BUILD	CHN 24%
SC-SEMIS	Power semiconductors	Components	2.5%	demand 0.32	6.2%	low 0.35	BUILD	CHN 32%
SC-SEMIS	Integrated circuits	Components	3.8%	low n/a	5.2%	low n/a	LOW INTERVENTION	CHN 16%
SC-SEMIS	Power electronics components	Components	2.8%	high 0.73	8.6%	low 0.62	BUILD	CHN 37%
SC-SEMIS	Automatic controllers	Assembly	15.6%	low 0.28	11.3%	low (cell 0.67; not a material buyer)	LOW INTERVENTION	CHN 12%
SC-SEMIS	Static converters & inverters	Assembly	4.3%	demand 0.28	4.2%	low (cell 0.67; not a material buyer)	BUILD	CHN 38%
SC-SOLAR	Polysilicon	Refining	30.7%	low (cell 0.71; not a material buyer)	7.0%	high 0.95	LOW INTERVENTION	CHN 97%
SC-SOLAR	Wafers	Refining	9.9%	low 0.21	28.4%	low 0.32	BUILD	CHN 98%
SC-SOLAR	Silver powder	Refining	25.8%	low n/a	59.1%	low n/a	LOW INTERVENTION	0%
SC-SOLAR	Solar cells	Components	0.9%	demand 0.21	0.3%	low 0.32	BUILD	CHN 92%
SC-SOLAR	PV cover glass	Components	2.5%	demand 0.21	0.2%	low 0.32	BUILD	CHN 61%
SC-SOLAR	PV modules & generators	Assembly	4.5%	demand 0.20	0.1%	demand 0.55	BUILD	CHN 84%
SC-WIND	Blade core wood (balsa)	Refining	0.1%	demand n/a	0.0%	not measured n/a	LOW INTERVENTION	CHN 3%
SC-WIND	Wind blades & hubs	Components	8.2%	demand 0.27	2.8%	low 0.60	BUILD	CHN 71%
SC-WIND	Wind towers	Components	0.4%	demand 0.27	0.5%	low 0.60	LOW INTERVENTION	CHN 18%
SC-WIND	Hydro turbine parts	Components	2.0%	demand n/a	0.6%	demand n/a	COORDINATE	CHN 35%
SC-WIND	Wind turbine generating sets	Assembly	7.3%	low 0.33	0.0%	high 0.75	BUILD	CHN 74%
SC-WIND	Hydro turbines	Assembly	3.1%	low n/a	0.3%	low n/a	LOW INTERVENTION	CHN 21%

Sources: 1.

Table A2. Section 3, Bilateral Complementarity Map

RANK	TIER	TECHNOLOGY	UNIT	STAGE	US POSITION	JAPAN POSITION	DYADIC RELATIONSHIP	STRATEGIC CASE
1	joint bet	Solar Manufacturing	Polysilicon	Refining	strong (30.7% of world supply)	moderate (7.0% of world supply)	US supplies / Japan needs	geopolitical chokepoint; need 0.71 / 0.95
2	joint bet	Electric Vehicles	Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds)	Refining	moderate	moderate	Shared vulnerability	geopolitical chokepoint; need 0.90 / 1.00
3	joint bet	Wind	Wind refining (copper wire & rod, rare-earth metals, rare-earth oxides & compounds)	Refining	moderate	weak	Shared vulnerability	geopolitical chokepoint; need 0.96 / 0.69
4	joint bet	Permanent Magnets & Drive Systems	Rare-earth oxides & compounds	Refining	moderate (5.3% of world supply)	moderate (6.3% of world supply)	Shared vulnerability	geopolitical chokepoint; need 0.96 / 1.00
5	joint bet	Electric Vehicles	Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite)	Extraction	moderate	weak	Capacity-constrained trusted supply	geopolitical chokepoint; need 0.82 / 0.98
6	joint bet	Batteries	Batteries extraction (cobalt ores, lithium minerals, natural graphite)	Extraction	weak	weak	Shared vulnerability	geopolitical chokepoint; need 0.81 / 0.98
7	joint bet	Battery Materials & Cells	Natural graphite	Extraction	moderate (3.1% of world supply)	moderate (1.7% of world supply)	Shared vulnerability	geopolitical chokepoint; need 0.82 / 0.98
8	joint bet	Wind & Hydropower Equipment	Wind turbine generating sets	Assembly	moderate (7.3% of world supply)	weak (0.0% of world supply)	Capacity-constrained trusted supply	geopolitical chokepoint; need 0.33 / 0.75
9	joint bet	Batteries	Batteries components (accumulator parts & separators)	Components	strong	moderate	US supplies / Japan needs	strategic manufacturing gap; need 0.06 / 0.75
10	joint bet	Battery Materials & Cells	Accumulator parts & separators	Components	moderate (3.1% of world supply)	strong (11.0% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need 0.73 / 0.75

RANK	TIER	TECHNOLOGY	UNIT	STAGE	US POSITION	JAPAN POSITION	DYADIC RELATIONSHIP	STRATEGIC CASE
11	joint bet	Permanent Magnets & Drive Systems	Rare-earth permanent magnets	Components	moderate (1.8% of world supply)	strong (8.8% of world supply)	Capacity-constrained trusted supply	geopolitical chokepoint; need 0.73 / 0.62
12	joint bet	Oil	Oil extraction	Extraction	strong	weak	Capacity-constrained trusted supply	trusted partner capacity constraint; need 0.47 / 0.98
13	coordinate	Solar Manufacturing	Wafers	Refining	moderate (9.9% of world supply)	strong (28.4% of world supply)	Complementary specialisation	geopolitical chokepoint; need 0.21 / 0.32
14	coordinate	Solar	Solar refining (copper wire & rod, polysilicon)	Refining	weak	moderate	Shared vulnerability	geopolitical chokepoint; need 0.71 / 0.95
15	coordinate	Wind	Wind assembly (electricity meters, overhead conductors, power cables)	Assembly	strong	moderate	US supplies / Japan needs	strategic manufacturing gap; need 0.33 / 0.75
16	coordinate	Battery Materials & Cells	Battery materials (foil, electrolyte)	Refining	moderate (2.8% of world supply)	strong (11.2% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need 0.45 / 0.94
17	coordinate	Battery Materials & Cells	Manganese dioxide (EMD)	Refining	moderate (3.7% of world supply)	strong (18.3% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need 0.45 / 0.94
18	coordinate	Battery Materials & Cells	Battery recycling feed	Recovery	strong (43.7% of world supply)	moderate (1.6% of world supply)	Complementary specialisation	emerging bottleneck; need 0.88 / 0.98
19	coordinate	Batteries	Batteries assembly (li-ion cells & packs, other accumulators)	Assembly	strong	moderate	US supplies / Japan needs	geopolitical chokepoint; need 0.90 / 0.71
20	coordinate	Semiconductors & Power Electronics	Piezo-electric quartz	Refining	moderate (3.3% of world supply)	strong (10.8% of world supply)	Capacity-constrained trusted supply	geopolitical chokepoint; need n/a / n/a
21	coordinate	Gas	Gas assembly	Assembly	moderate	strong	Complementary specialisation	strategic manufacturing gap; need 0.18 / 0.40
22	coordinate	Gas	Gas extraction	Extraction	strong	weak	US supplies / Japan needs	no material intervention; need 0.30 / 0.94
23	coordinate	Gas	Gas refining	Refining	strong	weak	US supplies / Japan needs	no material intervention; need 0.09 / 0.74
24	coordinate	Electric Grid	Electric Grid refining (copper smelting & refining, copper wire & rod, grain-oriented electrical steel (goes))	Refining	weak	moderate	Shared vulnerability	geopolitical chokepoint; need 0.90 / 0.71
25	coordinate	Heat Pumps & Cooling	Refrigerants	Refining	strong (19.9% of world supply)	weak (1.4% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need n/a / n/a
26	coordinate	Industrial Electrification & Hydrogen	Electrolysers	Assembly	moderate (7.5% of world supply)	strong (13.4% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need n/a / n/a
27	coordinate	Heat Pumps & Cooling	Refrigeration & heat-pump compressors	Components	moderate (7.6% of world supply)	strong (11.2% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need n/a / n/a
28	coordinate	Semiconductors & Power Electronics	Sensors & controller parts	Components	strong (10.1% of world supply)	moderate (8.6% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need n/a / n/a
29	coordinate	Industrial Electrification & Hydrogen	Furnace parts	Components	moderate (7.7% of world supply)	strong (11.0% of world supply)	Capacity-constrained trusted supply	trusted partner capacity constraint; need n/a / n/a
30	coordinate	Permanent Magnets & Drive Systems	AC generators	Assembly	strong (27.4% of world supply)	weak (1.5% of world supply)	Complementary specialisation	strategic manufacturing gap; need 0.27 / 0.60
31	coordinate	Oil	Oil assembly (ac generators)	Assembly	moderate	strong	Complementary specialisation	strategic manufacturing gap; need 0.14 / 0.36
32	coordinate	Coal	Coal components	Components	moderate	strong	Complementary specialisation	strategic manufacturing gap; need 0.49 / 0.71
33	coordinate	Coal	Coal assembly	Assembly	moderate	strong	Complementary specialisation	strategic manufacturing gap; need 0.42 / 0.59

RANK	TIER	TECHNOLOGY	UNIT	STAGE	US POSITION	JAPAN POSITION	DYADIC RELATIONSHIP	STRATEGIC CASE
34	coordinate	Oil	Oil refining	Refining	strong	moderate	Complementary specialisation	strategic manufacturing gap; need 0.00 / 0.53
35	coordinate	Coal	Coal extraction	Extraction	strong	weak	US supplies / Japan needs	no material intervention; need 0.61 / 0.97
36	manage competition	Gas	Gas components	Components	strong	strong	Direct competition	no material intervention; need 0.03 / 0.59
37	manage competition	Semiconductors & Power Electronics	Automatic controllers	Assembly	strong (15.6% of world supply)	strong (11.3% of world supply)	Direct competition	no material intervention; need 0.28 / 0.67
38	manage competition	Semiconductors & Power Electronics	Semiconductor manufacturing equipment	Assembly	strong (15.6% of world supply)	strong (21.2% of world supply)	Direct competition	no material intervention; need 0.20 / 0.55
39	manage competition	Solar Manufacturing	Silver powder	Refining	strong (25.8% of world supply)	strong (59.1% of world supply)	Direct competition	no material intervention; need n/a / n/a

Sources: 1.

Table A3. Section 4, Trade Agenda

ID	DIRECTION	UNIT	ACTION TYPE	ACTION
T01	USA -> JPN	polysilicon	predictability	Keep US exports of Polysilicon to Japan tariff-free and predictable; the buyer's need is met by the trusted supplier.
T02	both	Electric Vehicles/Refining	anti circumvention	Common origin and anti-circumvention rules for Electric Vehicles refining (rare-earth metals, rare-earth oxides & compounds): both import heavily from CHN, so a measure one side takes alone leaks through the other.
T03	both	Wind/Refining	common standards	Common specifications for Wind refining (copper wire & rod, rare-earth metals, rare-earth oxides & compounds) so a joint supplier base can serve both markets.
T04	both	rare-earth-oxides-and-compounds	anti circumvention	Common origin and anti-circumvention rules for Rare-earth oxides & compounds: both import heavily from CHN, so a measure one side takes alone leaks through the other.
T05	USA -> JPN	Electric Vehicles/Extraction	procurement recognition	Recognise US output of Electric Vehicles extraction (cobalt ores, lithium minerals, natural graphite) as trusted domestic-equivalent supply in Japan procurement and content rules; capacity, not access, is the constraint.
T06	both	Batteries/Extraction	common standards	Common specifications for Batteries extraction (cobalt ores, lithium minerals, natural graphite) so a joint supplier base can serve both markets.
T07	both	natural-graphite	common standards	Common specifications for Natural graphite so a joint supplier base can serve both markets.
T08	USA -> JPN	wind-turbine-generating-sets	procurement recognition	Recognise US output of Wind turbine generating sets as trusted domestic-equivalent supply in Japan procurement and content rules; capacity, not access, is the constraint.
T09	USA -> JPN	Batteries/Components	predictability	Keep US exports of Batteries components (accumulator parts & separators) to Japan tariff-free and predictable; the buyer's need is met by the trusted supplier.
T10	JPN -> USA	accumulator-parts-and-separators	procurement recognition	Recognise Japan output of Accumulator parts & separators as trusted domestic-equivalent supply in US procurement and content rules; capacity, not access, is the constraint.
T11	JPN -> USA	rare-earth-permanent-magnets	procurement recognition	Recognise Japan output of Rare-earth permanent magnets as trusted domestic-equivalent supply in US procurement and content rules; capacity, not access, is the constraint.
T12	USA -> JPN	Oil/Extraction	procurement recognition	Recognise US output of Oil extraction as trusted domestic-equivalent supply in Japan procurement and content rules; capacity, not access, is the constraint.
T13	open	uranium-fuel-upstream--aggregated	open sourcing	Leave Uranium/fuel Extraction: aggregated low-coverage sub-sectors (Spent fuel & nuclear scrap, Uranium ores) open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T14	open	pgms-iridium-ruthenium	open sourcing	Leave PGMs (iridium, ruthenium) open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T15	open	nuclear-special-materials	open sourcing	Leave Nuclear special materials open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T16	open	transformers	open sourcing	Leave Transformers open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T17	open	uranium-conversion-and-enrichment	open sourcing	Leave Uranium conversion & enrichment open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T18	open	copper-ores-and-concentrates	open sourcing	Leave Copper ores & concentrates open to global sourcing: trusted supply is adequate and diversified, so a bilateral preference adds cost without security.
T19	both	Gas/Components	no preference	No bilateral preference for Gas components: both are strong suppliers; keep the market contestable.
T20	both	automatic-controllers	no preference	No bilateral preference for Automatic controllers: both are strong suppliers; keep the market contestable.
T21	both	semiconductor-manufacturing-equipment	no preference	No bilateral preference for Semiconductor manufacturing equipment: both are strong suppliers; keep the market contestable.

ID	DIRECTION	UNIT	ACTION TYPE	ACTION
T22	both	silver-powder	no preference	No bilateral preference for Silver powder: both are strong suppliers; keep the market contestable.

Sources: 1.

Rule-generated from the data (R/partnerships) with external facts from a dated, curated evidence layer; not yet checked by an independent verifier. Items marked partial or unverified must be confirmed before citation.

Allied Industrial Base Explorer, Partnership memo: United States–Japan Strategic Energy & Industrial Partnership